

K-Means Clustering of Spotify Audio Features

Beats Analytica · Spring 2026 · Unsupervised Genre Analysis

1. Overview

This analysis applies unsupervised K-means clustering to 4,830 Spotify tracks using ten measurable audio features. The goal is not to build a genre classifier but to ask a more fundamental question: **do tracks that sound alike also share a genre label?** Discrepancies between the audio-derived clusters and Spotify's playlist genre labels reveal where genre is a sonic construct and where it is a cultural or marketing one.

4,830

Tracks analyzed

10

Audio features used

k = 4

Clusters selected

21.3%

Overall genre match rate

Source datasets

high_popularity_spotify_data.csv + low_popularity_spotify_data.csv

Algorithm

K-means — `kmeans()` , `nstart = 50`, `iter.max = 100`, `set.seed(123)`

Audio features

danceability, energy, tempo, valence, acousticness, instrumentalness, speechiness, liveness, loudness, duration_ms

Cluster selection

k = 4 (WSS elbow + silhouette plateau)

Evaluation

Cluster-to-genre majority label match rate

Script

deliverables/kmeans_analysis.R

2. Model Selection — Choosing k

Candidate values k = 2 through 8 were evaluated on two metrics: average silhouette score (measures how well each track fits its assigned cluster relative to its nearest neighbor cluster) and total within-cluster sum of squares (WSS, measures compactness).

k	Total WSS	Between SS	Avg Silhouette
2	35,086	13,204	0.341
3	31,580	16,710	0.204
4	28,437	19,853	0.199 Selected
5	26,023	22,267	0.193
6	24,052	24,238	0.168
7	22,560	25,730	0.169
8	21,097	27,193	0.175

Critical reading of the metrics

Silhouette peaks at $k = 2$ (0.341) and drops sharply at $k = 3$ (0.204), after which it plateaus. Two clusters would collapse the dataset into "acoustic/quiet" vs. "energetic/produced" — too coarse to be analytically useful. The plateau from $k = 4$ onward means additional clusters do not impose meaningfully greater separation costs. **$k = 4$ was selected** at the WSS elbow where four interpretable sonic archetypes emerge without over-segmentation. The moderate overall silhouette scores (0.17–0.34) confirm that audio features do not partition the musical landscape into sharply separated groups — genre overlap in audio space is measurable and real.

3. The Four Cluster Profiles

All ten features were z-score scaled before clustering. Feature means below are in original units.

Cluster 1 — Quiet Acoustic n = 355 · 7.3%

Dominant: classical (24.8%) · Top 3: classical, wellness, ambient

Acousticness	0.93
Instrumentalness	0.84
Loudness	−29.7 dB
Energy	0.07
Valence	0.09
Speechiness	0.04

Near-zero energy, almost entirely instrumental, exceptionally quiet. Classical, wellness, and ambient converge here because all three share the same audio signature despite being culturally distinct. Even within the cluster, the dominant genre captures only 24.8% — the remainder reflects context-driven labels (yoga playlist, game soundtrack) rather than sound.

Cluster 2 — Vocal Rhythmic n = 789 · 16.3%

Dominant: hip-hop (30.0%) · Top 3: hip-hop, latin, ambient

Speechiness	0.30
Danceability	0.72
Energy	0.66
Valence	0.57
Acousticness	0.26
Instrumentalness	0.03

Speechiness at 0.30 is nearly 5× any other cluster — the defining feature. Hip-hop and latin share this space because both rely on vocal delivery and percussive rhythm, despite cultural difference. Ambient tracks with spoken-word or vocal layers are pulled in by speechiness scores.

Cluster 3 — Mellow Semi-Instrumental n = 991 · 20.5%	
Dominant: lofi (26.8%) · Top 3: lofi, jazz, electronic	
Acousticness	0.66
Instrumentalness	0.51
Energy	0.37
Valence	0.34
Loudness	−12.0 dB
Speechiness	0.06
The "chill" audio space. Quieter and more organic than the mainstream clusters, yet more present than C1. The 99 electronic tracks here are downtempo/ambient electronic — not high-BPM productions. Low valence (0.34) signals an introspective, moody tendency across lofi, jazz, and ambient electronic.	

Cluster 4 — Energetic Mainstream n = 2,695 · 55.8%	
Dominant: pop (16.2%) · Top 3: pop, electronic, rock	
Energy	0.72
Danceability	0.66
Valence	0.56
Acousticness	0.17
Instrumentalness	0.05
Loudness	−6.4 dB
Holds 55.8% of all tracks. The dominant share is only 16.2% — the lowest of any cluster — because this cluster represents a production aesthetic, not one genre. Pop, rock, electronic, and latin all share the energetic, commercially produced sound. Genre labels within this cluster are cultural markers, not acoustic descriptions.	

4. Genre Distribution across Clusters

Track counts for ten key genres across all four clusters. Concentration reflects what share of a genre's tracks fall in its primary cluster.

Genre	C1 Quiet Acoustic	C2 Vocal Rhythmic	C3 Mellow Semi-Instr.	C4 Energetic Mainstream	Total	Concentration
ambient	73	103	58	125	359	Low
classical	88	0	30	3	121	High
electronic	67	54	99	369	589	Low
hip-hop	0	237	7	151	395	Medium
jazz	2	6	116	22	146	High
latin	0	111	47	267	425	Low
lofi	19	11	266	2	298	High
pop	0	23	56	436	515	Medium
rock	0	5	18	322	345	Medium
wellness	80	0	0	0	80	High

Strong sonic identity (High concentration)

Classical, lofi, wellness, and jazz concentrate in one cluster because their audio profile is consistent and measurable. The label describes a sound.

Weak sonic identity (Low concentration)

Ambient, electronic, and latin scatter across three or more clusters. These labels describe mood, geography, or cultural context — not a stable set of audio properties.

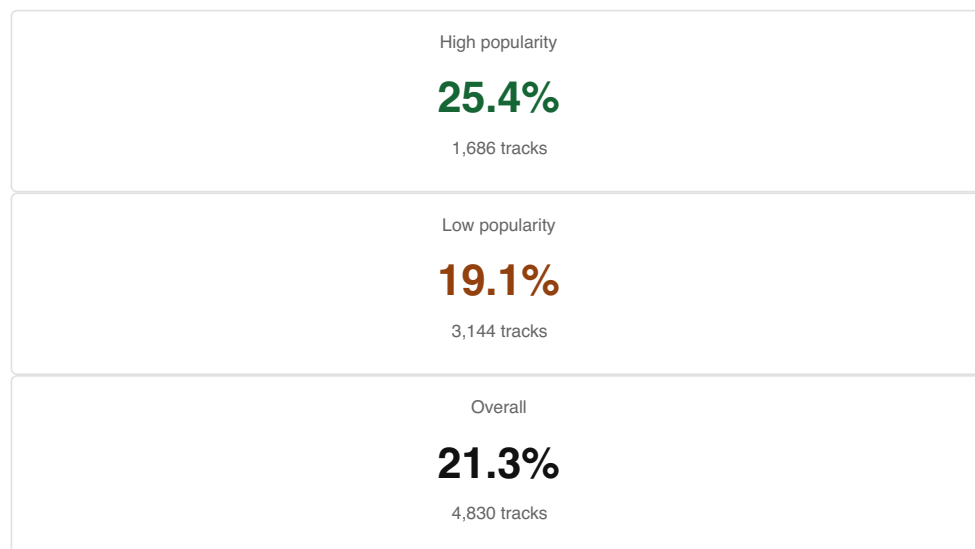
5. Genre Match Rate Analysis

5.1 Overall: 21.3%

Each cluster is assigned its majority genre label. Applying that label to every track in the cluster and comparing it to the track's actual Spotify genre yields a 21.3% agreement rate. The critical interpretation: **this number is the finding, not a measure of model quality.**

78.7% of tracks are assigned to a cluster whose sonic archetype differs from their genre label. These are not mislabeled tracks — they are evidence that genre definitions operate on non-audio dimensions. Genres where the label *is* an audio description (classical, lofi) would score high in isolation; genres that are cultural or commercial categories (ambient, electronic, latin, pop) drag the overall rate down.

5.2 By Popularity Tier



High-popularity tracks align with their genre labels ~33% more often. This gap is explained by dataset composition, not by any inherent property of popularity itself.

Genre	High popularity (top 5)	Low popularity (top 5)
1st	pop (357)	electronic (441)
2nd	rock (235)	ambient (298)
3rd	hip-hop (227)	lofi (296)
4th	latin (184)	latin (241)
5th	electronic (148)	world (224)

The high popularity dataset is dominated by genres with well-established, commercially reinforced sonic identities. The low popularity dataset includes far more culturally defined genres (ambient, world, lofi, brazilian, arabic) whose labels carry less audio information.

5.3 Match Rate by Cluster and Popularity Tier

Cluster (majority label)	High match rate	Low match rate	Gap
C1 — hip-hop	42.9%	23.2%	+19.7 pp
C2 — classical	64.3%	23.2%	+41.1 pp
C3 — lofi	1.2%	32.2%	−31.0 pp
C4 — pop	24.4%	9.3%	+15.1 pp

Classical cluster, high popularity — 64.3%

Only 14 high-popularity tracks land in the classical cluster; almost all are genuinely classical recordings. High-popularity classical music is acoustically unmistakable. Low-popularity tracks from wellness, ambient, and gaming migrate into this cluster because they share the audio profile but not the cultural label.

Lofi cluster, high popularity — 1.2%

172 high-popularity tracks land in the lofi cluster — meaning they *sound* like lofi — but almost none carry the lofi genre label. High-popularity music is almost never labeled lofi regardless of its audio properties. This is one of the clearest demonstrations in the dataset of the gap between sonic and cultural genre assignment.

Lofi cluster, low popularity — 32.2%

Low-popularity tracks in this cluster are the genre's natural home: 266 of 991 Cluster 3 tracks are genuinely labeled lofi. The cluster's highest single-genre match rate across all low-popularity cluster pairings.

6. Critical Assessment

What the model does well

- Recovers four meaningful sonic archetypes from unlabeled audio data with clear feature-level interpretability.
- Identifies which genres have a stable, measurable audio identity (classical, lofi, wellness) vs. which are culturally constructed categories (ambient, electronic, latin).
- The popularity-tier split reveals a structural asymmetry: genre-audio coherence is stronger for mainstream, commercially reinforced genres than for niche or culturally specific ones.

What the model cannot do

- **K-means assumes spherical clusters.** Real genre distributions are likely non-convex — genres like "ambient" or "electronic" may form multiple disconnected regions in audio space that K-means collapses into one assignment.
- **The majority-label match rate is a blunt metric.** A track labeled "latin" that lands in the hip-hop cluster may be a reggaeton production acoustically indistinguishable from hip-hop — the mismatch is informative, not erroneous.
- **No temporal or artist-level features are used.** Genre is partly a function of who makes the music and when — two tracks with identical audio features may be categorized differently based on artist identity or release era.
- **Cluster labeling is arbitrary.** The majority-genre label assigned to each cluster changes if dataset composition changes — as seen in the popularity-split re-run, where cluster numbering shifted. The audio archetypes are stable; their majority-genre labels are artifacts of dataset balance.
- **k = 4 is an analytical choice, not ground truth.** With 35+ genre labels in the dataset, four clusters can only approximate the structure. Higher k captures more variation but reduces interpretability.

Implications for the broader project

The K-means results establish a baseline: **audio features alone explain roughly 21% of genre label assignment**. The remaining 79% is driven by factors outside the audio signal. This creates a principled benchmark against which the supervised KNN classifier and the LLM few-shot classifier can be compared:

- If KNN substantially outperforms 21.3%, the additional signal comes from learning decision boundaries from labeled examples rather than from any inherently better audio representation.
- If the LLM outperforms both, it is drawing on world knowledge about genre conventions that neither audio features nor labels alone can capture.
- The popularity-tier finding motivates a potential segmentation strategy: genre prediction may be more tractable for high-popularity mainstream tracks and inherently less reliable for the long tail of niche genres.

7. Output Files

All model outputs are in `deliverables/kmeans_outputs/`.

File	Contents
<code>k_selection_metrics.csv</code>	WSS, between-SS, and silhouette for k = 2–8
<code>cluster_feature_means.csv</code>	Mean value of each audio feature per cluster
<code>cluster_sizes.csv</code>	Track count per cluster
<code>cluster_genre_summary.csv</code>	Dominant genre, share, and top-3 genres per cluster
<code>genre_by_cluster_counts.csv</code>	Full genre × cluster cross-tabulation (35+ genres)
<code>track_level_cluster_assignments.csv</code>	Per-track cluster, predicted genre, and match flag
<code>kmeans_pca_clusters.png</code>	PCA scatter plot (PC1 vs PC2) colored by cluster
<code>kmeans_summary.txt</code>	Console-formatted summary of all key results

Analysis performed in R · `deliverables/kmeans_analysis.R` · Model: `kmeans()` , nstart = 50, k = 4, set.seed(123) ·
Evaluation: `cluster::silhouette()` · Beats Analytica, April 2026